

client_udp.py > ...

```
1  import socket
2
3  def start_client():
4      host = "127.0.0.1"    # server IP
5      port = 5000          # same port as server
6
7      client_socket = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
8
9      message = input("Enter message to send to server: ")
10     client_socket.sendto(message.encode(), (host, port))
11
12     client_socket.close()
13
14 if __name__ == "__main__":
15     start_client()
```

server_udp.py > ...

```
1  import socket
2
3  def start_server():
4      host = "127.0.0.1"    # localhost
5      port = 5000          # use any free port
6
7      server_socket = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
8      server_socket.bind((host, port))
9
10     print(f"UDP Server listening on {host}:{port}...")
11
12     data, addr = server_socket.recvfrom(1024)
13     print("Message received from client:", data.decode())
14     print("Client address:", addr)
15
16     server_socket.close()
17
18 if __name__ == "__main__":
19     start_server()
```



```
PS D:\VS\CN> python server_udp.py
UDP Server listening on 127.0.0.1:5000...
Message received from client: hello server
Client address: ('127.0.0.1', 52918)
```

```
PS D:\VS\CN> python client_udp.py
Enter message to send to server: hello server
```

client_tcp.py > ...

```
1  import socket
2
3  def start_client():
4      host = "127.0.0.1"    # server IP
5      port = 5000          # same port as server
6
7      client_socket = socket.socket()
8      client_socket.connect((host, port))
9
10     message = input("Enter message to send to server: ")
11     client_socket.send(message.encode())
12
13     client_socket.close()
14
15  if __name__ == "__main__":
16     start_client()
```

server_tcp.py > ...

```
1  import socket
2
3  def start_server():
4      host = "127.0.0.1"
5      port = 5000
6
7      server_socket = socket.socket()
8      server_socket.bind((host, port))
9      server_socket.listen(1)
10
11     print(f"Server listening on {host}:{port}...")
12
13     conn, addr = server_socket.accept()
14     print("Connection from:", addr)
15
16     data = conn.recv(1024).decode()
17     print("Message received from client:", data)
18
19     conn.close()
20     server_socket.close()
21
22  if __name__ == "__main__":
23     start_server()
24
```

```
PS D:\VS\CN> python server_tcp.py
Server listening on 127.0.0.1:5000...
Connection from: ('127.0.0.1', 53144)
Message received from client: hello server
```

```
PS D:\VS\CN> python client_tcp.py
Enter message to send to server: hello server
```